

Sushant Raj

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Work Experience

Locus Robotics

Senior Robotics Integration Engineer

August 2025 - Present

- Contribute to research and development of prototype robots aimed at improving efficiency in logistics and fulfillment applications
- Develop and integrate core systems across the robot software stack
- Support experimentation with novel hardware/software integrations to expand robotic capabilities and enable new fulfillment workflows

Robotics Software Engineer (Perception Team)

July 2023 - August 2025

- Supported integration/filtering of diverse sensors and accessories (lidars, depth, RGB cameras, conveyors, etc), enhancing robot perception capabilities and enabling different use cases for each of the different Locus Robots
- Created tools to easily diagnose sensor hardware issues remotely to streamline debugging processes, reducing the need for onsite resources to diagnose issues
- Supported development and bring-up for new robot platforms (Vector 1.2, Vector 2.0 and Array)
- Collaborated with other teams to support interfacing the perception pipeline with other robot components, such as planning & controls
- Created a system for model performance tracking & testing for the Perception team's ML models in order to provide insight into the performance of the ML features over time
- Developed and implemented an improved depth camera ground segmentation pipeline to enhance obstacle detection and avoidance
- Implemented tools for tracking sensor filter pipeline performance and latencies and used performance analysis tools (such as VTune) to find and improve inefficiencies in the system
- Implemented a new lightweight layer to help robots avoid tall obstacles that slip out of the sensors' field of view

Robotics Software Engineer

June 2022 - July 2023

- Supported various new deployments (remotely and on site) of Locus Vectors into customer sites and developed new functions to integrate the Vector system with Waypoint Dispatcher interface into the existing Locus deployment infrastructure.
- Supported developing and deploying tools for new use cases, such as cart detection systems and robots with fixed carts.
- Collaborated on creating an embedded system tester used by Locus' contract manufacturer, was responsible for developing the software used to communicate with the embedded system board and run tests to verify functionality.
- Worked on improvements to the 3D lidar filtering in order to detect and avoid smaller obstacles

Robotics Engineering Intern

October 2021 - June 2022

- Helped integrate a manipulator onto the existing robot platform. This involved creating functions to command the manipulator from the existing robot's UI
- Created a standalone application for visualization of robots using the Mass Robotics Interoperability Standard and implemented support on Waypoint robots

Waypoint Robotics

July 2018 - Oct 2021

Robotics Intern

- Utilized MongoDB to streamline the usage of additional products such as wireless charging docks, alignment markers, and carts
- Managed and fulfilled feature requests to fix bugs and add functions to the user interface
- Created a web-based visualization application to display sensor data
- Completed projects that consisted of creating scheduling functions for the robot, developing a torque-based safety feature, and GPS integration

Education

Worcester Polytechnic Institute

May 2026

Master of Science, *Robotics Engineering*

University of Connecticut, Honors College, STEM Scholar

May 2022

Bachelor of Science, *Electrical Engineering & Computer Engineering & Minor: Mathematics*

Achievements

- First Place at HackUConn Hackathon 2021
 - Dean's List (Fall 2021)
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Skills

Technical: Proficiency in Python, C++, GDB, OpenVino, ROS, Git, VSCode, VTune, Grafana

Languages: Fluent in English, Tamil, and Sanskrit

Leadership and Involvement

University of Connecticut Indian Student Association

May 2019 - May 2022

Directed a board of 14 talented individuals, planning several events for over 1200 attendees, and serving as a leader of the South Asian community. Served as Media Chair, Chief Financial Officer, and President.

MetroHacks

MetroHacks was a nonprofit organization dedicated to hosting hackathons for high school students. In addition to providing an event where students could compete to create STEM projects to fulfill different challenges, the organization aimed to foster an environment for learning and growth by hosting workshops and organizing mentors to help guide participants at the events.

Board Member

January 2019 – May 2021

- Expanded MetroHacks to three new locations including New York City, Atlanta, Washington DC
- Remotely managed and advised the local teams on financial, marketing, and branding/design initiatives for each event

Director of Branding & Sponsorship

Fall 2017 – August 2018

- Pursued sponsorship from companies to support the organization and created branding materials for events